

PowerLine Command (PLC) - Asteroid Build Report

Version 1.5 - Asteroid Command Theme with Music

Executive Summary

Project: PowerLine Command Training System
Version: v1.5Ast (Asteroid Command)
Build Type: Single HTML File (Self-Contained)
File Size: 11 MB
Total Lines: 120,361 lines
Build Date: January 2026
Theme: Space/Asteroid Command

Technical Architecture

Build Composition

File Structure:

- Single HTML file with embedded assets
- Inline CSS styling (space theme)
- Inline JavaScript game engine
- Embedded base64 background music (MP3)
- No external dependencies

Technologies Used:

- HTML5
 - CSS3 with CSS Variables and Animations
 - Vanilla JavaScript (ES6+)
 - Canvas-free DOM-based rendering
 - Base64 audio embedding
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Dataset Configuration

PowerLine Training Dataset (3-File Formula)

The game uses a modular three-file approach to generate training challenges:

File 1: Command Types (3 types)

- **AP** - Assign
- **ENP** - Enroute
- **BSEH** - Base Unit

File 2: Units (106 units)

Emergency response unit codes including:

- 2100-2126 series (27 units)
- 2040-2046 series (7 units)
- 2095-2099 series (5 units)
- 2133-2139 series (8 units)
- 2150 (1 unit)
- 2200-2208 series (6 units)
- 2302-2339 series (7 units)
- 2345-2346 series (2 units)
- 2360-2369 series (10 units)
- 2391-2398 series (8 units)
- 2520-2523 series (4 units)
- 2720-2725 series (6 units)
- 2B01-2B02 series (2 units)
- Special units: FIT, MHRT (2 units)
- CARE1-CARE8 series (8 units)

File 3: Base Stations (21 bases)

Niagara Region locations with station codes:

Station Name	Code
Niagara Falls	72100
Ontario St	72101
Linwell	72102
Thorold	72103
NOTL	72104
Grimsby	72105
Port Colborne	72107
Welland	72108
Smithville	72109
Vineland	72110
Pelham	72111
Ridgeway	72113
Glendale	72115
St Paul	72116
Fort Erie	72117

Station Name	Code
Merittville	72118
HQ	72120
Fitch St	72121
Westwood	72122
Fleet	72123
Fallsview	72124

Dataset Statistics

Total Possible Combinations: 6,678

- Command Types: 3
- Units: 106
- Bases: 21
- Formula: $3 \times 106 \times 21 = 6,678$ unique training scenarios

Game Mechanics

Core Configuration

```
BASE_FALL_DURATION: 25,000ms (25 seconds)
BASE_SPAWN_DELAY: 2,500ms (2.5 seconds)
DIFFICULTY_INCREASE_RATE: 0.05 (5% per correct answer)
SPAWN_REDUCTION_RATE: 0.04 (4% faster spawning)
MAX_SIMULTANEOUS_TARGETS: 3 (base) + bonus targets
MAX_SHIELDS: 3 lives
```

Scoring System

Points Distribution:

- **Correct Answer:** 10 points
- **Perfect Timing Bonus:** +5 points
- **Streak Bonuses:**
 - 5-hit streak: +25 points
 - 10-hit streak: +50 points
 - 10+ streak: +50 points per 10 hits
- **Multi-kill Bonus:** +15 points

Progressive Difficulty

The game implements dynamic difficulty scaling:

- Difficulty multiplier increases by 5% per correct answer
- Spawn delay decreases progressively
- Maximum simultaneous targets increases (1 per 10 correct answers)
- Fall speed increases with difficulty

Gameplay Features

Challenge Format:

- Display: [Command Type] [Unit] [Base Name]
- Example: "Assign 2100 Niagara Falls"

Required Input:

- Format: [CMD] [UNIT] [CODE]
- Example: "AP 2100 72100"

Performance Tracking:

- Real-time accuracy percentage
 - Streak counter
 - Shield/life system
 - Perfect timing detection (no backspace usage)
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Visual Design

Theme: Asteroid Command (Space)

Color Scheme:

- Background Dark: #0a0a15
- Background Darker: #000000
- Primary Text: #00ffff (Cyan)
- Secondary Text: #8888ff (Light Blue)
- Target Color: #666666 (Gray)
- Challenge Color: #ffff00 (Yellow)
- Command Color: #0088ff (Blue)
- Zone Color: #ff0066 (Pink)

Visual Effects:

- Neon glow effects with text shadows
- Animated falling asteroids (targets)
- Particle explosions on correct hits
- Ground impact effects on misses
- Screen shake on impacts
- Gradient zone barrier
- Bonus notification popups
- Pause overlay effects

UI Components

Screens:

1. Start Screen (Mission Select)
2. Options Screen (Module Selection)
3. Game Screen (Active Gameplay)
4. Pause Screen
5. Game Over Screen (Mission Complete)

HUD Elements:

- Score counter (top-left)
 - Accuracy percentage (top-left)
 - Shields/lives indicator (top-right)
 - Command input panel (bottom-center)
 - Bonus notifications (right side)
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Audio Integration

Background Music

Specifications:

- Format: MP3
- Encoding: Base64 embedded
- Title: "Asteroid Command"
- Source: Suno AI Generated
- Creator: darque_shadow592
- Size: ~10.8 MB (embedded)
- Auto-play: Enabled (with fallback for blocked autoplay)

Audio Controls:

- Starts automatically on game launch
 - Loops continuously
 - Stops on menu return
 - Restarts on replay
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Control Scheme

Keyboard Controls

Menu Navigation:

- **1** - Select primary option
- **2** - Select secondary option (main menu)
- **ESC** - Back/Exit to menu

Gameplay:

- **Type** - Enter command
- **ENTER** - Submit command
- **ESC** - Pause game
- **SPACE** - Resume (when paused)

Tracking:

- Backspace usage tracked for perfect score detection
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Performance Metrics

Statistics Tracked

Game Session:

- Final Score
- Accuracy Percentage
- Perfect Hits (no backspace)
- Total Targets Destroyed
- Current Streak
- Shields Remaining

Real-time Updates:

- Score
 - Accuracy
 - Shield count
 - Bonus notifications
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Build Features

Optimization

Code Structure:

- Modular function organization
- Event-driven architecture
- Efficient DOM manipulation
- Cleanup routines for destroyed elements
- Interval-based game loop

Asset Management:

- Single-file deployment
- No external requests (except allowed network access)
- Base64 audio embedding
- CSS animations (GPU-accelerated)

Browser Compatibility

Requirements:

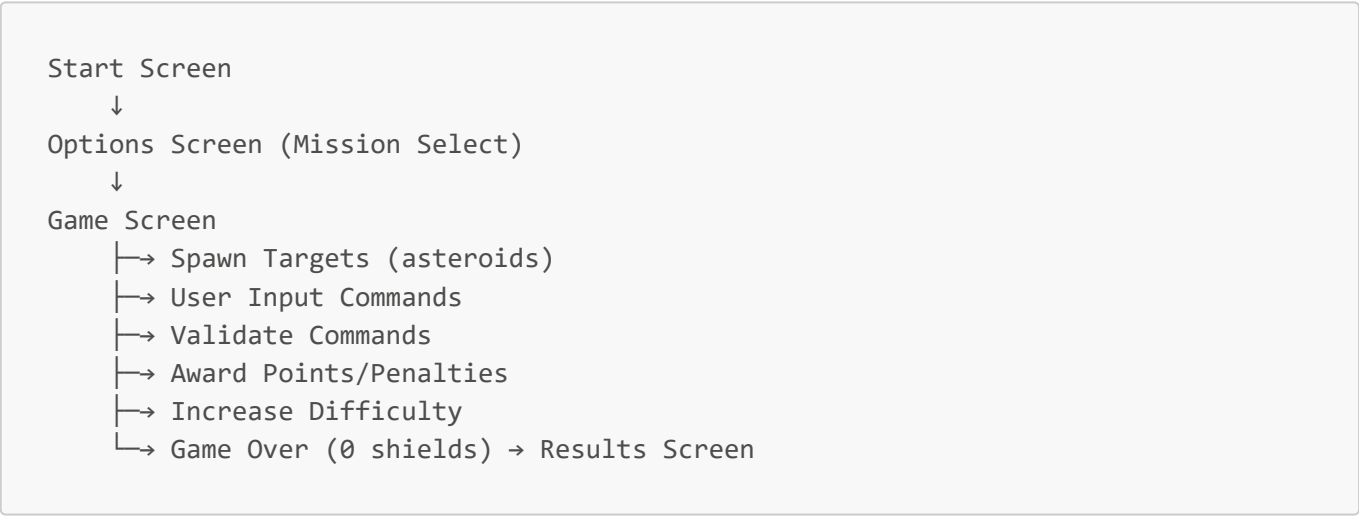
- Modern browser with ES6 support
- Audio playback capability
- DOM manipulation support
- CSS3 animation support

Tested Features:

- Keyboard event handling
- Input focus management
- Audio autoplay (with fallback)
- Screen transitions
- Real-time updates

Gameplay Loop

Flow Diagram



Target Lifecycle

1. **Spawn:** Random challenge generated
2. **Display:** Falls from top of screen
3. **Validation:** User enters command
4. **Result:**
 - Correct: Explosion effect, points awarded
 - Incorrect/Timeout: Ground impact, shield lost
5. **Cleanup:** Element removed from DOM

Data Generation Algorithm

Challenge Generation

```
function generateChallenge() {  
  const cmdType = random(commandTypes)    // 1 of 3  
  const unit = random(units)              // 1 of 106  
  const base = random(bases)              // 1 of 21  
  
  return {  
    challenge: `${cmdType.challenge} ${unit} ${base.challenge}`,  
    command: `${cmdType.command} ${unit} ${base.command}`  
  }  
}
```

Randomization: True random selection ensures varied training scenarios

Build Quality Assessment

Strengths

- ✓ **Self-Contained:** Single HTML file deployment
- ✓ **Complete Dataset:** 6,678 training combinations
- ✓ **Progressive Difficulty:** Adaptive challenge scaling
- ✓ **Rich Feedback:** Visual and audio effects
- ✓ **Performance Tracking:** Comprehensive statistics
- ✓ **Professional Theme:** Polished space aesthetic
- ✓ **Embedded Audio:** Custom background music included

Technical Achievements

- ✓ **No Dependencies:** Pure vanilla JavaScript
 - ✓ **Efficient Rendering:** DOM-based without canvas overhead
 - ✓ **Smooth Animations:** CSS3 GPU acceleration
 - ✓ **Event-Driven:** Clean architecture
 - ✓ **Modular Data:** Three-file formula approach
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Deployment Information

Format: Single HTML file

Deployment: Direct browser open (file://) or web server

Requirements: Modern web browser only

Installation: None required

Updates: Replace single file

Version History

v1.5Ast - Asteroid Command (Current)

- Space theme implementation

- Embedded background music
- 3-file formula dataset
- 21 Niagara Region bases
- 106 emergency units
- 3 command types
- 6,678 total training combinations

Base: v1.5 Roman Numerals Demo

Skin: Asteroid Command (Space Theme)

Dataset: PowerLine Commands (3-file formula)

Recommendations

Future Enhancements

1. **Difficulty Modes:** Easy/Medium/Hard presets
2. **Leaderboard:** Local storage high scores
3. **Training Mode:** Slower pace with hints
4. **Statistics Dashboard:** Detailed performance analytics
5. **Custom Playlists:** Additional music tracks
6. **Mobile Support:** Touch controls
7. **Save Progress:** Session persistence

Educational Use

Ideal For:

- Emergency dispatch training
- Command code memorization
- Response time improvement
- Accuracy training
- Stress management practice

Training Benefits:

- Gamified learning
 - Immediate feedback
 - Progressive difficulty
 - Performance metrics
 - Engaging presentation
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Conclusion

PLC v1.5 Asteroid Command is a fully-featured, self-contained training game that combines educational value with engaging gameplay. The build successfully implements a comprehensive dataset of 6,678 training scenarios across 3 command types, 106 units, and 21 base stations. The space theme provides an immersive experience while the progressive difficulty ensures appropriate challenge scaling.

Build Status: ✓ Complete and Functional

Quality: Professional

Deployment Ready: Yes

Report Generated: January 2026

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